8 · *Ecological Scales: Issues of Resolution and Extent*

Scale is an important issue in ecological applications; many patterns emerge only at a specific scale, and are perceived as noise or constants, or even go undetected at other scales (Chave, 2013). When preparing the data, one common problem is that the datasets we intend to use do not always match in terms of scale or resolution. We therefore need to scale spatial data to a common spatial extent and resolution in order to obtain a consistent modeling environment (see e.g. Section 6.2 for examples). This is definitely the easiest approach, but there are also other approaches, when a hierarchy of scales is included and implemented in a habitat suitability modeling exercise (see e.g. Lomba et al., 2010; Gallien et al., 2012; Fernandes et al., 2014).

Scaling is an important aspect of modeling, and it is therefore important to recall its conceptual foundation. Scaling is basically the process of translating information from one scale to another (King, 1991; Wu, 1999, 2004). A prerequisite for scaling models is usually to first adopt a concept of how the processes and components involved interrelate and how they can be ordered. Hierarchy theory provides a conceptual framework for this ordering (O'Neill et al., 1986; Allen and Hoekstra, 1992; Schneider, 2001). Depending on the nature of the processes or patterns, scaling can be a complex undertaking (Lischke et al., 2007). When carrying out habitat suitability and species distribution modeling the scaling is comparably simple. There are basically two important aspects to take into consideration: the effect of resolution and the effect of extent.

When considering resolution in a spatial context, we immediately think of "spatial resolution." For a grid, this naturally refers to its cell size. However, for the same grid it can also mean to what extent the thematic information (the map content) is resolved (e.g. how many and to what extent habitat units are distinguished or what units for temperature are used), or to what extent the map resolves temporal information. For

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our purposes, we will therefore distinguish three major levels of resolution: spatial, temporal, and thematic. Scaling often involves either the choice of a specific resolution, or to scale the data to appropriate or common resolution, by up- or downscaling. While some of these latter processes can be quite tricky, we only use simple examples that can be easily implemented in a GIS/statistical framework (see Chapter 6).

The extent scale can also be analysed on the three basic levels: spatial, temporal, and thematic. Extent effects are more challenging to deal with when modeling the suitability of habitats. What is the appropriate extent for solving a problem? Answering this question is by no means straightforward. On the one hand, we can easily say that once we work in a study area, then the spatial extent is defined. In many instances, however, the questions we ask cannot be answered from within the extent of the study area alone. If our study area is comparably small and encompasses relatively little topographic variation, then habitat suitability modeling under strong climate change scenarios by 2100 will drive most of the species' potential habitats out of the study area (and we can be fairly certain about this aspect), but we lack any certainty about what other species may actually find suitable habitats in this same study area by 2100. This is simply because these species cannot be observed within the current extent. Not even for those existing within the study area, we can be sure to fit all their niche limits appropriately, as they likely extend far beyond the study area. In such cases, HSMs do not reflect the full environmental constraint that limits their distribution.

In the following section, we will discuss how far questions of resolution and extent are relevant in the preparation, analysis and modeling of spatial ecological data.

8.1 Issues of Resolution

8.1.1 Spatial Resolution

Basically, all studies modeling habitat suitability and species distributions, encounter the problem of spatial resolution, both at the level of the species distribution data (dependent variable when modeling) and at the level of spatial predictors. We will briefly evaluate the techniques and principles relevant to the spatial resolution of our data.

Species data usually come either as point measurements or as raster cells from atlas data. The data may be classified as presence–absence or come in any abundance, frequency, proportion or importance scale. When available as raster information – the *Atlas Florae Europaeae* (Jalas

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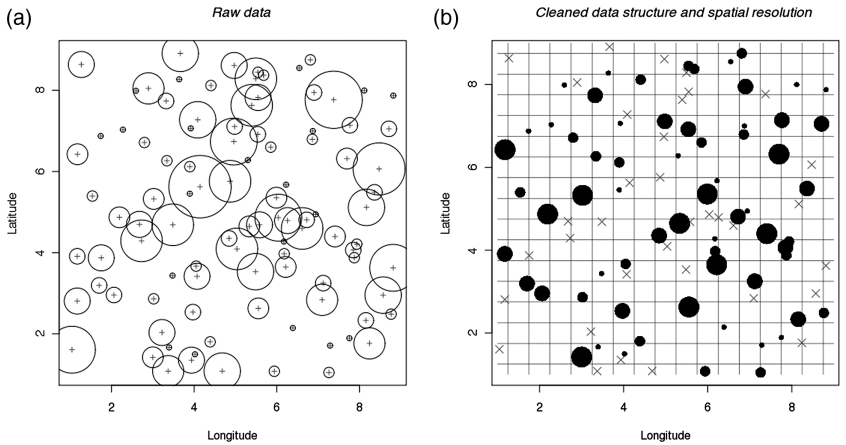


Figure 8.1 (a) Species occurrence data with positional uncertainty illustrated by circles around the noted location (+) mapped along artificial "latitude" and "longitude" coordinates. (b) Remaining points with accepted positional uncertainty and the associated spatial resolution of the raster (gray lines) used for calibration and prediction. The excluded species occurrences are shown with a red cross.

and Suominen, 1972–1996) is one example – then the spatial resolution is defined by the nature of the data (e.g. 50 km). When data are available as point (or small plot) level information (e.g. GBIF data or sampling plots), then the required spatial resolution is not immediately clear. A single point observation is conceptually infinitely small or as small as the size of the plant or animal of concern. However, we cannot assume that the point coordinates are accurate with no error. How large is this error? Some coordinates may be as imprecise as ±1 km or even only be available at the resolution of 0.1°. It makes a huge difference for our subsequent analyses if the coordinates are highly precise (errors < 5 m) or highly imprecise (errors > 10 km). In the latter case, it is difficult to relate our species information to topographic or environmental site variations, since the low precision means the information from a high-resolution surface of environmental predictors, such as climate or topography, cannot be retrieved accurately from the point coordinates. We therefore need to ascertain that the spatial resolution used at the predictor variable level fits with the uncertainty of the position coordinates for the point samples. A virtual example is given in Figure 8.1. Assume varying positional accuracy for the coordinates at each point (illustrated as circles around the position center in panel a). Since we do not want to work with overly high levels of uncertainty, we may want to

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remove a set of points where the positional uncertainty exceeds a given threshold. In our artificial example, we set the raster resolution to a value of `0.25` (indicated by gray lines in panel b) and remove all the points with greater spatial uncertainty (indicated by the red crosses in panel b). Alternatively, a coarser resolution for elevation could be used with an increased number of points. However, this may not increase the accuracy of the species–environment relationship, but rather introduce a bias, since coarser resolutions blur the environmental variability contained in a higher resolution grid.

There are no objective rules for making decisions on the exclusion or inclusion of points with regards to their positional accuracy. It seems important to reflect on the aim of the analysis before taking such decisions. How much topographic variation and smaller-scale ecological gradient is required for our analysis? Our decision needs to consider these questions in combination with the consequences on sample size when excluding points (see Section 7.3). The result is essentially a trade-off between keeping as many points as possible to increase the sample size, and removing as many points as possible to increase the spatial accuracy, the spatial resolution, and with this the accuracy of the species–environment relationship in the habitat suitability model.

While such decisions are relatively easy to make when the positional accuracy is known, it is more difficult to select a subset of useful data when no or only limited information on positional accuracy is available. If we have additional information at hand for given points, such as elevation, slope, or aspect, then this would be sufficient to at least use digital topography to test which points actually fit the field observed topography. In this case, we ignore the question of whether the field derived positional accuracy or the positional and topographic accuracies of the DEM we use actually cause more serious problems. Our primary aim is to ascertain that the field and digital topography match. We can test for this by overlaying each point in a GIS with DEM-derived elevation, slope and aspect. Most points will show slightly different values for these three variables. Again, we need to define what errors we will accept in our modeling. Imagine that we have a point where the field protocol says it is a steep south-facing slope, and the GIS database says it is a moderately steep north-facing slope. Whatever the reason for this mismatch, we would obviously attach all the environmental variables relating to north-facing slopes to the point in our database, while in fact we know that the point, and in particular the growth conditions for the species we have observed at this location, are significantly different. By deciding how

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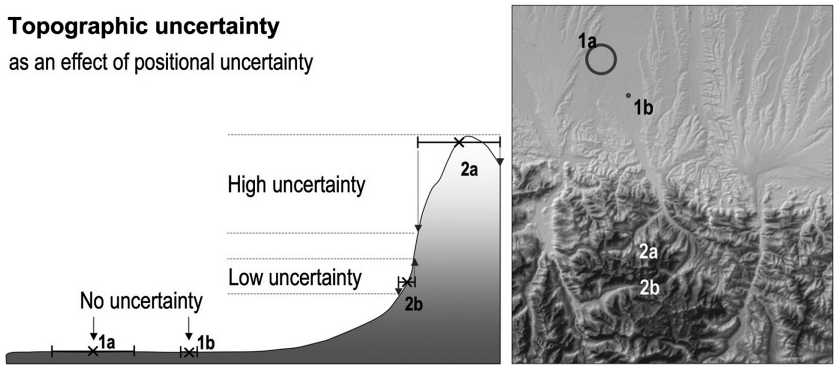


Figure 8.2 Relationship between positional and topographic uncertainty. Positional uncertainty matters especially where there are strong topographic and/or environmental gradients (2a/2b), and do not significantly influence our modeling where there are no such steep gradients visible (1a/1b). (*A black and white version of this figure will appear in some formats. For the color version, please refer to the plate section.*)

much topographic error is acceptable, we decide how well we are able to differentiate between different topographic situations. In essence, this relates to the thematic resolution (Section 8.1.3), as it determines how well we can actually differentiate along topographic axes. Another positive effect of this method for optimizing spatial scale effects is that we can accept larger positional errors in topographically homogenous terrain, since even comparably large positional uncertainties will not affect the linking of GIS-related "ecology." Figure 8.2 illustrates the link between positional and topographic uncertainty.

Most predictor layers are now available as raster maps. One major inconsistency that needs to be addressed at the beginning of a modeling study is that all raster maps have the same spatial resolution, the same lower-left coordinate (alignment), and the same spatial extent. Section 6.2.3 provides example code for resampling and aligning raster layers so that they perfectly match with a master grid that serves as a reference for all mapping. Working with differing resolutions may cause uncontrolled interpolation effects, and these are not easy to deal with if not treated explicitly (McPherson et al., 2006).

Spatial resolution, both at the level of the dependent and independent variable, has an effect on model fit and accuracy. In fact, it impacts more than just accuracy. First, coarser resolution data do not necessarily result in poorer model fit and accuracy (Mitchell et al., 2001; Guisan et al., 2007a; Luoto et al., 2007). Some studies have shown an improvement in

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predictive accuracy when higher resolution variables are used (Lassueur et al., 2006), yet this is primarily the effect of substituting climate for topography variables, not a direct effect of the resolution per se. Another study demonstrated that when scale differences between local point observations and used raster cells of predictors are too large, the models become less accurate (Seo et al., 2009). Specifically, predictor variables with grain sizes of larger than 10 km fitted to point observations of species presence–absence caused model accuracy to decrease significantly in this study. In general, however, models that use dependent and independent coarse resolution data show very high model fit, with AUC values often above 0.9. However, and this is ecologically problematic, they do not meaningfully fit climate–species relationships, only relationships on very broadly averaged information. Most species are not uniformly distributed within large cells, and most large cells are not flat. This means that the true climate–species relationship is often obscured in coarse scale models. Despite coarse-grained data generally fitting HSMs successfully and producing very good test statistics, it remains rather uncertain which habitat drivers really are responsible for determining the distribution of species. In fact, it has been shown that coarse resolution HSMs tend to predict high extinction rates under projected future climates that do not match with projections when finer grained data are used (Randin et al., 2009). Such finer grained information obviously identifies suitable habitats that are smoothed away at the coarse grain.

8.1.2 Temporal Resolution

Temporal resolution is rarely properly considered in the context of modeling habitat suitability and species distributions. We usually collect a set of distribution data, and take "current" climate and environmental predictor variables to relate the distribution to these predictors. However, there might be a strong mismatch between the observed distribution and the drivers (predictors) responsible for shaping this distribution. We distinguish between three different aspects, which we discuss here briefly. The three aspects all relate to the problem of capturing the niche of a species when overlaying observed presence and absence points with climate and other environmental predictors and then fitting statistical models: (i) Does "current" climate refer to the period responsible for the observed presence of a species? (ii) Is there a time lag in a species' response to changes in environmental conditions or does the species respond instantaneously to these changes? And (iii) is it possible to

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observe all elements of a species' niche under current climate conditions or can different (non-analog) environments in the past also contributed to capturing a species' niche?

The first issue relates to the question of whether the time of observation fits the time of the "current" climate and environmental data as drivers of this distribution. If we have, say, observations from the period of 1970–1980, then basic climate normals data for the period of 1961–1990 might be perfectly fine as these cover the basic sampling period. However, this is not necessarily a sound assessment of all species and all circumstances. A species might be very long lived, as is the case for trees, and the climate which explains their distribution is likely to date from much earlier. This is especially true if we assume that the regeneration niche determines spatial distribution much more than the adult niche. We then face the problem that trees of between 80 and 150 years of age at the time of the aforementioned period would have been influenced by climate between 1820 and 1900. We can usually still fit the models successfully when using more recent climates. However, we might get a biased estimate of their climate niche if we only consider current climate to explain their distribution. Presence may rather be an expression of persistence than of a viable population niche for some tree species in some locations (Bell et al., 2014). This is not always easy to detect and relates back to the definition of the niche (Part I). While slow migration in response to changing environments may result in incomplete RF (see the discussion of the second issue), no migration or extremely slow migration may result in a current range pattern that cannot be explained well by current, but rather (and better) by past, climate and environmental conditions. Support for this view has, for example, been demonstrated for reptiles and amphibians in Europe, where the late glacial 0°C isotherm is a better predictor of narrow-range species than the current 0°C isotherm, whereas the latter better explains wide-range species (Araújo et al., 2008). In addition, Quaternary climate stability was found to better explain current species richness in reptiles and amphibians than current climate stability in the same study. Other species are very short lived, and respond very directly to fluctuating climate characteristics that are not correctly expressed with 30-year climate normals. Annual plants, marine phyto- or zooplankton all respond on a much faster timescale (resolution). While annuals primarily respond to spring and summer climates, and thus show high levels of annual fluctuation, marine plankton has an even shorter response time, basically from a few weeks to a few months at best. Here, each observation has to be related to the environmental

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(marine) conditions for that precise moment (week, month). Otherwise, the temporal resolution does not match. The best way of coping with this problem is to use primarily ecological and biological reasoning when selecting climate and environmental data for model-fitting purposes. This means that we need to identify the proper temporal resolution (what level of temporal detail) and extent (how far back in time, what time-span) of our predictors when planning a study. In fact, testing several alternative datasets may help to explore the best solution for a given dataset. Nowadays, there are large numbers of different climate datasets available spanning across larger time periods.

The second issue relates to the question of whether species actually track changing climate and environmental conditions successfully and steadily across the whole range. While fast dispersing species may follow this (pseudo-) equilibrium assumption (see Part I), many other species do not. There is ample evidence in the literature that there have been time lags in the re-adjustment to Quaternary climate change, for example. This evidence comes from the paleo-literature (Davis, 1989), as well as from species modeling studies (Svenning and Skov, 2004, 2007) and from recent observations (Gehrig-Fasel et al., 2007; Delzon et al., 2013; Lenoir and Svenning, 2015). Current climate change seems to shift with such fast velocities that many species are struggling to re-adjust (Foden et al., 2007). However, this may not yet have resulted in significant discrepancies between potential and actual ranges.

Another issue is the fact that many species do not fill their ranges everywhere they would find suitable habitats. This is primarily thought to originate from historical effects such as incomplete migration since the last Ice Age (Svenning and Skov, 2007) or regional changes in competitive interactions after environmental changes due to different migration and adaptation patterns in different species (Lenoir et al., 2010). For example, some studies have shown that the same trees that show limited RF actually seem to fill their niche dimension even across large, disjunct ranges (Randin et al., 2013). Although, filling is stronger at the cold (physiologically constraining) than at the warm limit, where the latter is thought to be more driven by biotic interactions than by physiological constraints (Meier et al., 2010, 2011). All this points to a general issue that may cause significant bias to our sampling and data, even if we perfectly design an environmentally sound sampling scheme. It may (and does) turn out that even if we properly sample gradients following a sound design, we might sample areas that have unfilled ranges for a given species. We then tend to infer these areas as a proof of unsuitable habitat

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for this species, when in fact it is not. Or we simply find significant noise in the calibration of the species–environment relationship, due to the fact that the species was not always there where the habitat was suitable. There is no best practice for coping with this problem. One solution is to test to what degree differently complex models do predict the observed range of a species, and whether over- or under-prediction rather only concern areas in the core or also at the edge of the range (Merow et al., 2014). Over-prediction in areas that represent the core of the ecological niche is likely to represent limited RF issues. Such an over-predicting, usually simple, model might then give relatively poor model accuracies in cross-validation or independent tests, but might actually be better than statistically evaluated, due to bias in the test data.

The third question has only recently been addressed. Many species, irrespective of whether they exhibit small or large ranges, do not find all of the environment they could tolerate under contemporary environmental conditions. This specifically concerns the interaction of the multiple environmental variables that, together, make up the species' niche. Former times may have exposed the species to different combinations of climates that have no modern analogs. This means that the contemporary range of a species may not give the full picture of its realized niche of a species. Maiorano et al. (2013) demonstrated that using climate and distribution data from the past 13 000 years revealed better niche model fits than data from a single 1000-year time slice. This suggests that the niche in this example was better captured with a multi-temporal analysis of the species' distribution. Such a model is likely to be much better suited to projecting the likely future habitat suitability of the same species under climate change than if it was fitted only from one time period. This is because the species will most certainly also face non-analog climates in the future, as was reported for the past (Williams and Jackson, 2007; Williams et al., 2007). As previously discussed, this third issue is a particular problem for projecting species through time, be it into the past or into the future. There is only limited experience to date on how to best cope with this problem. One way is to use either ensemble modeling techniques or to use models that are not overly complex. Ensemble modeling integrates statistical models of different complexities and different statistical properties (e.g. how much interaction among variables is considered or how a species' flexible responses along environmental gradients are fitted) when projecting a species through time (see Section 17.3). This ensures that several possible projections are considered. Conceptually, more complex models ensure more accurate fitting of a

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set of training data, but usually at the cost of overfitting and constraining the response too tightly to current correlations among predictor variables. Less complex models or calibrations tend to less accurately fit the models to observations, but tend to have improved transferability. This has been shown for space (Randin et al., 2006), but is also likely to hold for transferability through time (but see Araújo et al., 2005a).

8.1.3 Thematic Resolution

Thematic resolution is another scale issue that is rarely considered or discussed in habitat suitability and species distribution modeling studies. Thematic resolution concerns the extent to which a predictor variable resolves the thematic content it intends to represent. We can represent precipitation in units of millimeters, centimeters, decimeters, or meters, for example. While we may not be able to recognize the difference between millimeters and centimeters, this is clear for decimeters or meters when representing maps in these classified units. Representing precipitation as classified integer unit in meters would be considered a very coarse thematic resolution. Of course, we can always represent any climate variable as real rather than as integer numbers. This cannot, however, be done easily with all predictors, notably categorical variables. So we need to distinguish two issues that we will discuss in more detail below: (1) which thematic resolution is meaningful from an ecological viewpoint; and (2) which thematic resolution is meaningful from a statistical viewpoint?

The first question is not always easy to answer. Of course, we want to have high thematic resolution for all the predictors used. Regarding habitat units, geology, or soil information, we are often left with comparably coarse or ecologically uninformative classifications. A time-classified geologic stratification does not translate directly into ecologically meaningful classes. We therefore often have numerous classes that have more disadvantages than advantages for habitat suitability modeling, because we may not have sufficient observations of presence and absence for each of these strata. Here, we suggest reclassifying all strata that exhibit similar ecological properties, irrespective of age of the tectonic stratum. In this way, we generate a coarser temporal, but more a meaningful, thematic stratification. We can for example collapse sedimentary rock based on their clay and calcium content, which have direct impact on soil development, soil pH, and nutrient availability. In the end, we have a much smaller number of classes (thus lower thematic resolution), but these are more useful for modeling habitat suitability, and this also directly translates

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into easier sampling protocols and statistical modeling procedures (see the second point below). Generally speaking, we should always aim for a thematic resolution that is meaningful for the purposes of habitat suitability modeling. For example, representing precipitation in 0.1 mm may not be a good choice, since plants or animals do not directly respond to differences at this scale. Storing the maps in integer units of millimeters or centimeters will help in two ways: first, they are units that are more directly meaningful to organisms, and, second, it reduces the size of the stored map data. All categorical variables need to be reconsidered in the same way before starting a modeling exercise. If we have a habitat map available with, say, 50 classes then we need to ask ourselves: are all 50 classes important for explaining the spatial distribution of our target species, or can we safely lump some classes together (especially those that have no or a similar meaning for a target species). Simplifying our classification, not only has statistical benefits (see below), but we also generate a more parsimonious model, a model that is simpler, less complex, and easier to explain. Often, however, we face the opposite problem. We have only very coarse classes available, and we know from ecological reasoning that we should use a finer thematic resolution for our modeling exercise. In such cases, we need to decide if we make the effort to post-process the coarse categories and refine them from other sources, or if we keep them as they are, and then discuss the results in the light of this known limitation. The latter is often done where no (or insufficiently resolved) soil or land cover information is available. Refining classes requires considerable work. We can for example, refine an existing habitat classification (say five classes: forest, grassland, wetlands, unvegetated, urban) by means of ecological post-processing using either RS, or field data, or by simply classifying these classes into altitudinal belts or based on other prior knowledge. Reclassifying from RS data can be very tedious, while simple reclassifications from altitude or other forms of prior knowledge come with the risk of generating circular reasoning. Regarding numerical variables, we can consider the transformation of variables as an issue related to thematic resolution. For example, if we log-transform a precipitation gradient, then we add finer resolution to low values and coarser resolution to high values of precipitation. Such a transformation can be ecologically meaningful, since low precipitation values tend to have higher species turnover compared to high precipitation values.

The second question aims to answer how much thematic resolution is actually statistically meaningful. If we have answered the first question

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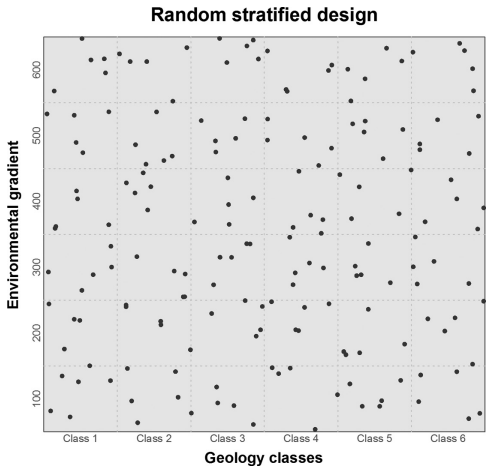


Figure 8.3 Illustration of a random stratified design along two variables, one being a nominal class (here geology) and one being a numerical environmental variable (here classified into classes of 100 units). The nominal variable has six classes, and the numerical gradient has also been classified into six classes. Within each class combination, a randomly allocated set of five observations has been chosen.

(on ecological reasoning), and if our sampling is perfect (sampling all thematic layers perfectly), then there are no real statistical limitations. However, we are usually confronted with thematic resolution problems after the fieldwork has been completed. From a statistical viewpoint, we need to have sufficient observations along each gradient (of numerical predictors) or within each thematic class (of nominal predictors). These are a prerequisite for building models that represent sound probabilistic responses of our target species to these variables. In fact, sufficient observations are needed for each possible combination of variables and categorical classes. Very often, we realize that we have no or only insufficient observations for some rare nominal classes (e.g. geological, soil, habitat units). In such cases, we cannot make meaningful predictions for these rare nominal classes. Similarly, when planning a stratified random sampling design we might ask ourselves if we will really be able to sample each gradient combination for each available nominal class variable (see Figure 8.3)? If our answer is no, then we need to recombine such rare classes with more frequent ones, preferably with classes that have similar ecological meaning for a species' distribution.

If, on the other hand, we have large numbers of classes available for a nominal predictor, then we first need to reconsider, which classes can

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actually be used in a statistically meaningful way. Similarly, as previously discussed, we may want to collapse several nominal classes that are too rare or that have very similar ecological meaning to other classes. This is not only important to reduce complexity for the sake of parsimony, but also has an impact from a statistical viewpoint. Each nominal class will use one degree of freedom in a regression-based model, so numerous nominal classes result in a large loss of degrees of freedom. We can only reduce this effect by collapsing nominal variables to fewer classes.

8.2 Issues of Extent

Extent is an important issue in habitat suitability and species distribution modeling. It primarily concerns the question: do my observation data represent the full or only a part of the full geographic range of my target species? Depending on the goal of the study, it can be very important to capture the full range of a species in order ascertain (at least under current climate conditions, see Section 8.1.2 for more discussion of this issue) if the full niche of a species is captured. Fitting models only using observation data from a subset of the whole range means it is not possible to fit the full niche, and it is likely that some important components of the niche will be overlooked. If the aim of the study is to locally predict habitat suitability under current environmental conditions (e.g. for conservation planning or species management purposes), then using only part of the range from local sampling at high spatial resolution is acceptable (Lyet et al., 2013). If, however, the goal is to project species habitat suitability to future climates, to test ecological hypotheses, or to project species habitat suitability to other, more distant, regions then, as far as possible, we need to capture the full niche of our target species (see Part I for discussion of this issue).

The effect of limited niche coverage on HSMs has been tested on numerous occasions, mostly in the context of climate change projections (Thuiller et al., 2004b; Barbet-Massin et al., 2010). These studies have shown that data from a subset of the current range do not represent all the conditions the species finds suitable throughout its whole range. Fitting a model to these incomplete datasets might cause high levels of bias when the models are projected through time or space (Thuiller et al., 2004b; Barbet-Massin et al., 2010). This limitation heavily constrains what the models fitted from such data can be used for. In one study that focused on evaluating the predictive performance of HSMs at the range margins of three widespread African vertebrates, the authors

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demonstrated that models fitted against fine (1 km) or coarse (10 km) resolution regional predictors performed better at the range margins than models fitted across a whole species range from coarse resolution (10 km) predictors (Vale et al., 2014). In another study, however, models fitted against coarse resolution (10 km) predictors capturing the whole Iberian range of amphibians predicted much better at regional scale than regionally fitted models, if the coarse resolution models were applied to high resolution (1 km) regional predictors (Suarez-Seoane et al., 2014). We have already discussed (Section 8.1.1) the fact that scaling from fine to coarse spatial resolution can obscure the real climate–species relationship (and thus the niche quantification). However, the example of Suarez-Seoane et al. (2014) suggests that applying a habitat suitability model fitted from coarse resolution predictors from the whole species range (thus representing a blurred niche shape) results in good performance at regional scales, due to the fact that it captures the whole niche and not only parts of it. This was, however, only true if this simple form of downscaling was applied. We argue that this is a valid way of scaling models fitted from the whole range to smaller regions. It is often applied even across continents, e.g. when fitting models at a coarser scale (e.g. 30' lat/lon), and then predicting at a finer scale (e.g. 10' lat/lon), done e.g. by Thuiller et al. (2005a). Indeed, it has been shown that this type of scaling is appropriate, as long as it does not exceed a ratio of 1:10 in scaling (Bombi and D'Amen, 2012). This approach is therefore a useful way of scaling models from the whole range of a species that can only be fitted from coarse resolution data to the finer resolution useful at regional scales.

If the focus is on regional predictions, then we often observe the phenomenon described by Vale et al. (2014, see above): range margins are partially poorly predicted from models fitted from the whole range, and regionally fitted models often perform better at these marginal locations. This may be for several reasons. On the one hand, the niche shape might be quite complex, and the training data might not always be sufficient to capture regional peculiarities in the niche that do not have sufficient probabilistic support from the whole range. On the other hand, such regional peculiarities can (and probably often do) represent adaptations to special ecological conditions at this range margin location or they might reflect different effects of dispersal and biotic interactions. Boulangeat et al. (2012a) developed a hierarchical habitat suitability model that could be used to test the differential effects of limited dispersal and biotic interactions on species distribution. This model allows

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for testing, to what degree the factors environment, dispersal or biotic interactions contribute to constraining the range margins of species. Of course, such margins can also represent locations where the range is insufficiently filled, which can also cause deviations from the margin conditions observed elsewhere. However, the local adaptation of populations across the full extent of a species' range is interesting both from a habitat suitability perspective and from range extent perspective, and it is difficult to comprehensively deal with. Several authors have started to study this effect, and they found quite surprising discrepancies between models fitted for infraspecific taxa (e.g. subspecies, genetic clades) that are then aggregated to the "whole species" and models fitted to the species data as a whole (Estrada-Peña and Thuiller, 2008; Pearman et al., 2010; D'Amen et al., 2013; Serra-Varela et al., 2015). From these studies, it becomes evident, that some infraspecific taxa or genetic clades: (i) show quite significant differences in niche structure and habitat preference, (ii) do not necessarily fill all the environmental space of the niche fitted from all intraspecific taxa jointly (species-level model), (iii) have only partially significant overlap in suitable habitats, and (iv) reveal obvious adaptations to more marginal environments than the species-level model. These results indicate that there may also be valid ecological reasons for fitting models from parts of the species' range only. If such a region is selected from genetic clustering, then fitting the models to observations from this region will capture the specific ecological niche of this genetic population, which may differ from that of populations from other parts of the niche. We expect to see more research in this area, bridging the gap between phylogeography and niche modeling in the coming years. Such studies will make an important contribution to better understanding the effects of extent and local adaptation.

Issue of temporal and thematic extents have partly been addressed already in relation to the treatment of temporal and thematic resolutions, and we therefore do not expand on these issues further.

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